REST API Documentation — Smart Irrigation System

Project: Smart Irrigation Alert Dashboard
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1. GET /api/sensors

- **Description:** Retrieve all registered sensor zones with current readings and status.
- **Request Headers:** `Accept: application/json`
- **Response Code:** `200 OK`
- **Sample Response:**

```
[
  {
    "id": 1,
    "zoneName": "Zone A - Wheat Field",
    "cropType": "Wheat",
    "moistureLevel": 22.5,
    "temperature": 31.0,
    "status": "CRITICAL",
    "lastUpdated": "2026-10-11T11:45:00"
  },
  {
    "id": 2,
    "zoneName": "Zone B - Orchard",
    "cropType": "Apple",
    "moistureLevel": 54.0,
    "temperature": 27.5,
    "status": "OPTIMAL",
    "lastUpdated": "2026-10-11T11:45:00"
  }
]
```

2. POST /api/sensors

- **Description:** Register a new sensor zone.
- **Request Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "zoneName": "Zone D - Rice Paddy",
  "cropType": "Rice",
  "moistureLevel": 45.0,
  "temperature": 29.0
}
```

- **Response Code:** `201 Created`

3. GET /api/alerts

- **Description:** Get all active critical exception zones (moisture $< 30\%$).
- **Response Code:** `200 OK`
- **Sample Response:**

```
{
  "totalAlerts": 1,
  "criticalThreshold": 30.0,
  "alerts": [
    {
      "zoneId": 1,
      "zoneName": "Zone A - Wheat Field",
      "currentMoisture": 22.5,
      "severity": "CRITICAL",
      "recommendation": "Activate Valve #1 for 45 minutes immediately."
    }
  ]
}
```

4. POST /api/simulate

- **Description:** Simulate real-time sensor fluctuation across all zones to test alert responsiveness.
- **Response Code:** `200 OK`
- **Response:**

```
{
  "message": "Simulation successful. Sensor data updated.",
  "timestamp": "2026-10-11T11:45:30"
}
```

5. POST /api/sensors/{id}/irrigate

- **Description:** Trigger an immediate irrigation cycle for a specific zone, raising its moisture to 75%.
- **Response Code:** `200 OK`